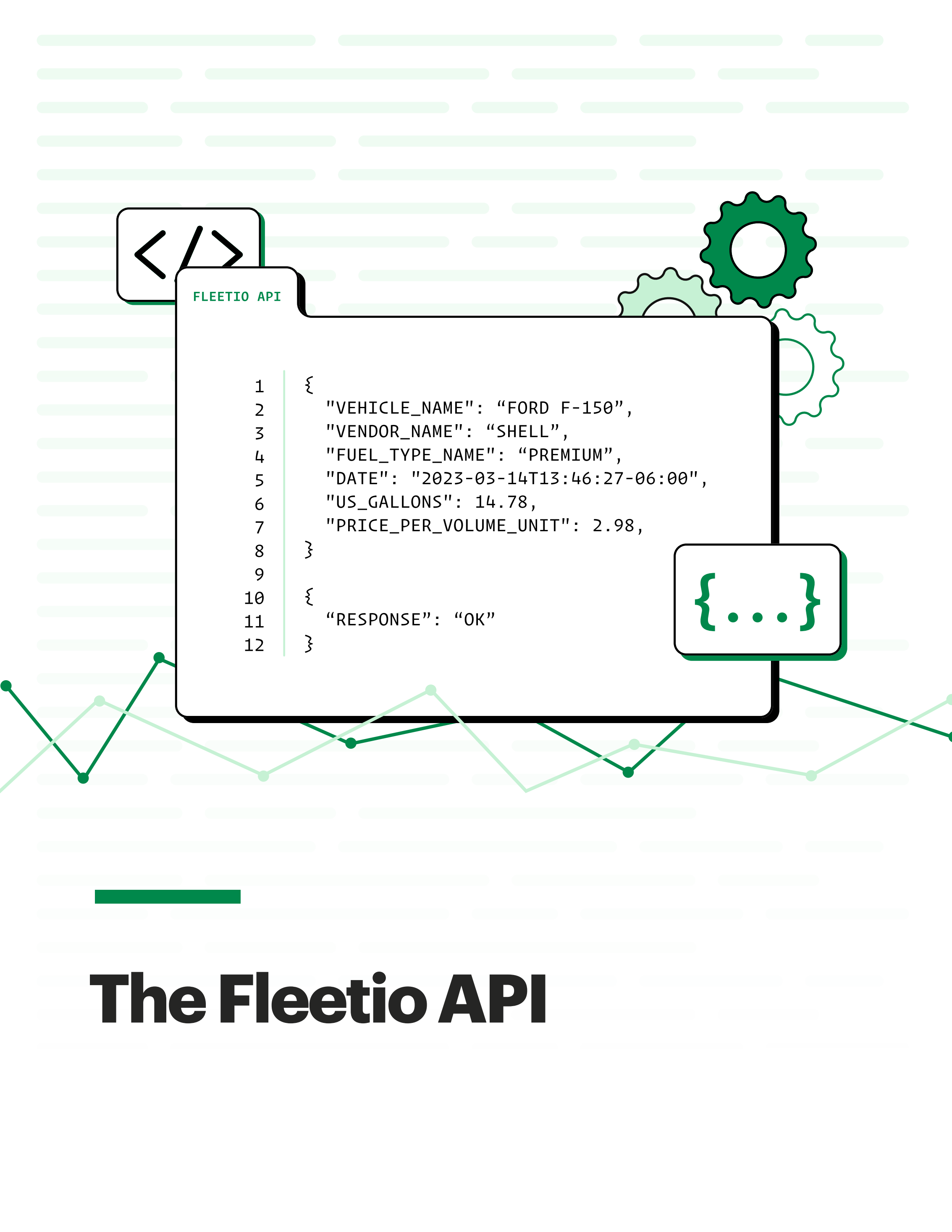




FLEETIO API

```
1  {
2    "VEHICLE_NAME": "FORD F-150",
3    "VENDOR_NAME": "SHELL",
4    "FUEL_TYPE_NAME": "PREMIUM",
5    "DATE": "2023-03-14T13:46:27-06:00",
6    "US_GALLONS": 14.78,
7    "PRICE_PER_VOLUME_UNIT": 2.98,
8  }
9
10 {
11   "RESPONSE": "OK"
12 }
```

{...}

The Fleetio API

The Fleetio API

Fleet management is an increasingly nuanced operation with a lot of complex needs when it comes to software. Fleets should never have to feel limited by their platform's capabilities – that's where APIs come into play. Fleetio offers a public API as a powerful solution to customize, streamline and optimize fleet operations through seamless integration and automation.

Whether you're a developer looking to integrate Fleetio with other systems or a fleet manager seeking to enhance operational efficiency, this guide will equip you with the knowledge and tools you need.

Here's What You'll Learn in the Guide

1

Get an in-depth understanding of Fleetio's API

2

Move confidently through setup, usage and best practices

3

See specific use cases for APIs in Fleetio

What is an API?

An API (Application Programming Interface) is a set of rules and protocols that allow different software applications to communicate with each other. It defines the methods and data formats that applications can use to request and exchange information.

Fleetio's API in particular is a set of tools and protocols that allow different software applications to communicate with Fleetio's fleet management system. It provides access to a wide range of data and functionalities, enabling you to automate workflows, integrate with other systems, and enhance your fleet management capabilities.

Why APIs Matter for Fleet Management

APIs help fleets stay in step with a rapidly-advancing industry by enabling seamless integration and automation. They provide fleet managers with the tools needed to streamline operations, improve efficiency and make data-driven decisions.

APIs play an important role in exchanging information between different systems and applications, which allows fleets to leverage all of the systems that drive their productivity so they can make better decisions, increase efficiency and reduce operational costs. With Fleetio's API, fleet managers can automate routine tasks, gain real-time insights and improve overall fleet performance.

Understanding Fleetio's API

How does Fleetio's API work?

Fleetio allows developers to access and manipulate fleet data programmatically via our REST API. It provides a standardized way to interact with Fleetio's fleet management system, and accomplish tasks like data retrieval, updates and workflow automation. Our API supports both public endpoints, ensuring secure and flexible access to your fleet data.

Benefits of Integrating with Fleetio's API

There are three things integrations should allow any fleet to do:

Avoid simple human errors
by automating data entry

Simplify repetitive day-
to-day tasks

See all of your data in
the correct context

Integrating with Fleetio's API presents a number of benefits, including:



Improved Efficiency: Automate routine tasks and reduce manual data entry.



Enhanced Decision-Making: Gain faster insights into fleet operations.



Customized Integration: Connect Fleetio with other software systems used in your organization, even if it's not supported by one of our integrations.



Scalability: Easily scale your fleet management operations as your business grows.

How Fleets Can Leverage APIs

Import Odometer Readings

Use Case:

Automatically update vehicle odometer readings from a telematics or GPS tracking system to ensure accurate mileage tracking.

Benefits:

- Ensures up-to-date and accurate odometer readings.
- Helps in scheduling timely maintenance based on actual vehicle usage.
- Reduces manual entry errors and saves time.

Example:

A telematics system sends odometer readings to Fleetio's Meter Entries API regularly. This keeps the vehicle mileage records current, triggering maintenance alerts when necessary.

Trigger Trouble Code Alerts

Use Case:

Import diagnostic trouble codes (DTCs) from a telematics system to proactively manage vehicle health and maintenance.

Benefits:

- Early detection of vehicle issues.
- Prevents minor issues from becoming major repairs.
- Improves vehicle uptime and reliability.

Example:

When a telematics system detects a trouble code, it sends the DTC to Fleetio's Faults API. Fleetio logs the code, triggers an alert and notifies the fleet manager for immediate action.

Update Fuel Transactions

Use Case:

For anything not covered by Fleetio's existing fuel card integrations, update fuel transactions from a fuel management system to monitor fuel usage and trigger location exceptions.

Benefits:

- Tracks fuel consumption accurately.
- Identifies fuel theft or unauthorized usage.
- Helps in optimizing fuel efficiency and reducing costs.

Example:

Fuel management system sends fuel transaction data to Fleetio's Fuel Entries API. Fleetio can log the transaction, check for location exceptions and alert the manager if any discrepancies are found.

Enhance Reporting & Analytics

Use Case:

Integrate with business intelligence tools to create comprehensive reports and analytics dashboards.

Benefits:

- Combines fleet data with other business metrics for holistic insights.
- Provides customizable and detailed reporting.
- Supports data-driven decision-making.

Example:

Fleetio's API allows extraction of vehicle, maintenance and operational data. This data can be fed into business intelligence tools like Tableau or Power BI to create detailed analytics dashboards, helping fleet managers visualize performance trends and make informed decisions.

Maintenance Scheduling & Alerts

Use Case:

Automate maintenance scheduling and alerts based on vehicle usage data.

Benefits:

- Ensures timely maintenance, reducing downtime.
- Extends vehicle lifespan.
- Improves overall fleet reliability and safety.

Example:

Fleetio can use vehicle data sent from telematics devices using our API in order to schedule maintenance. When a vehicle reaches a certain mileage or usage threshold, Fleetio can trigger a maintenance alert, notifying the fleet manager to schedule service.

But what if you're not technical?

That's okay! More than anything, APIs are simple logic – getting data from point A to point B without constant manual intervention. If the APIs you're working with are public, you can map out exactly what you need to happen and find someone with a little more code knowledge to help you on the technical side.

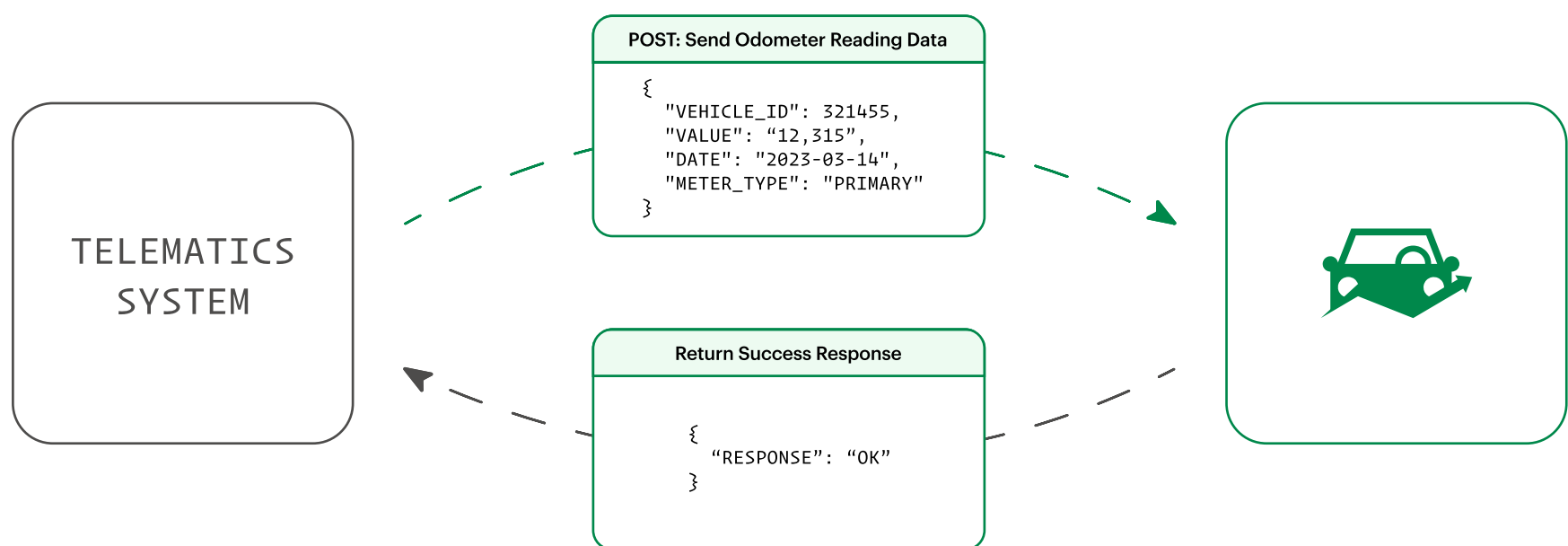
Tips for Integration and Customization

- **Plan Your Integration:** Define your integration goals and plan the API calls you need.
- **Prioritize Data:** Take advantage of filtering capabilities to tailor your range of data or narrow it to certain attributes.

Examples of API Maps

Import Odometer Readings

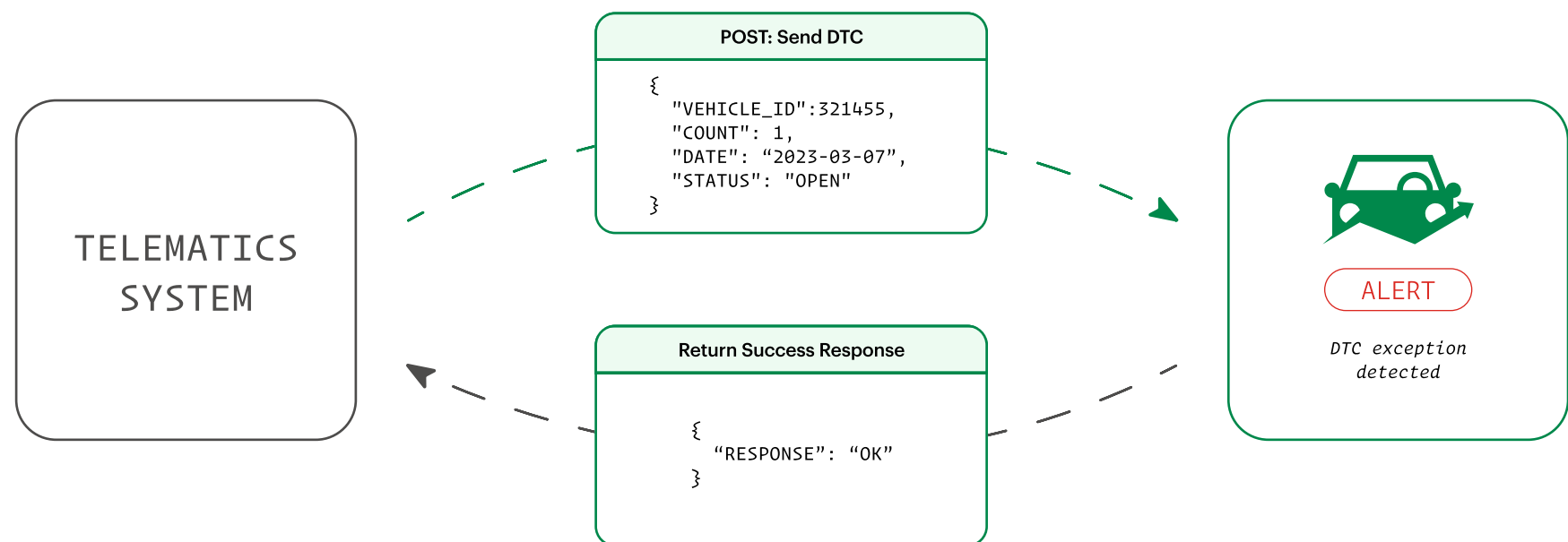
Purpose: Update meter readings from another system.



- 1 The telematics system sends a POST request to the endpoint `/api/v1/meter_entries` with the odometer reading data and associated vehicle ID.
- 2 Fleetio's API validates the request and updates the odometer reading in the database.
- 3 Fleetio's API returns a success response to the telematics system.

Trigger Trouble Code Alerts

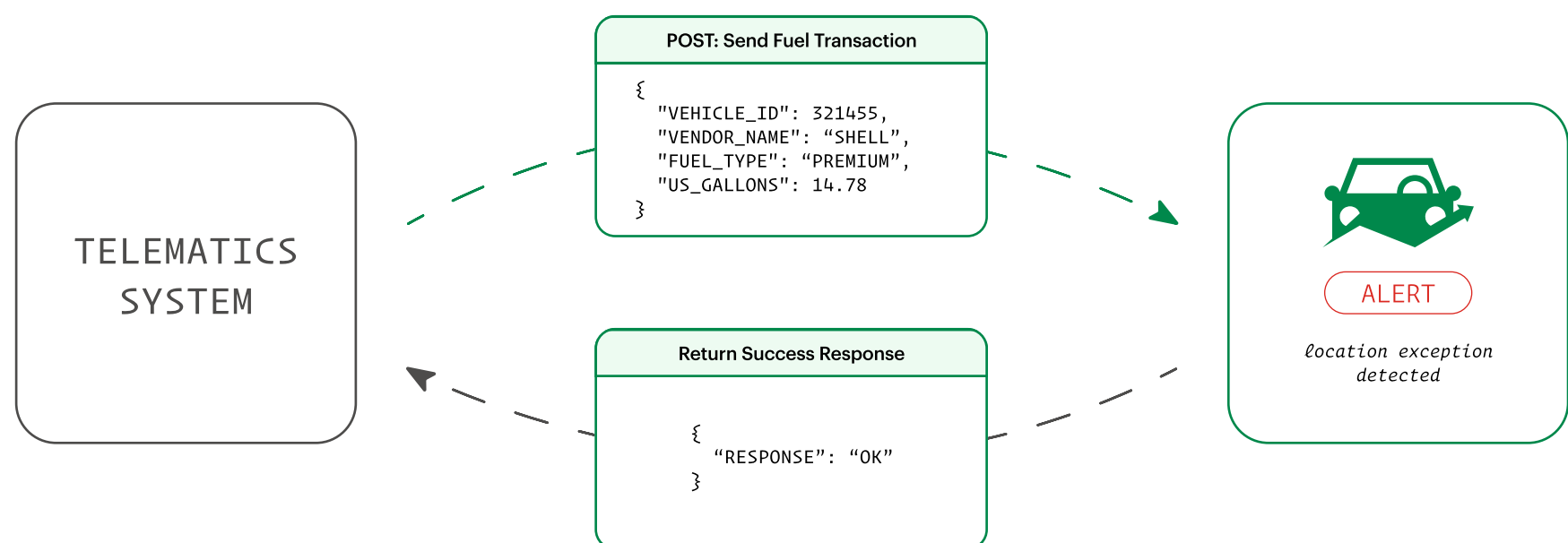
Purpose: Import diagnostic trouble codes (DTCs) from a telematics system.



- 1 The telematics system sends a POST request to the endpoint `/api/v1/faults` with the DTC data.
- 2 Fleetio's API validates the request and stores the DTC in the database.
- 3 Fleetio's API triggers an alert and notifies the fleet manager.
- 4 Fleetio's API returns a success response to the telematics system.

Update Fuel Transactions

Purpose: Update fuel transactions to trigger location exceptions.



- 1 The fuel management system sends a POST request to the endpoint `/api/v1/fuel_entries` with the fuel transaction data.
- 2 Fleetio's API validates the request and stores the fuel transaction in the database.
- 3 Fleetio's API checks for any location exceptions based on predefined logic.
- 4 If a location exception is detected, Fleetio's API triggers an alert and notifies the fleet manager.
- 5 Fleetio's API returns a success response to the fuel management system.

Best Practices for API Integration

1 Follow the Documentation Thoroughly

Read through [Fleetio's API documentation](#) carefully before starting your integration. The documentation provides detailed information on API endpoints, request and response formats, authentication, and error codes.

TIPS

- Keep the documentation handy for reference during development.
- Use the provided code samples and tutorials to understand common usage patterns.

2 Maintain Security

Security should be a top priority when integrating with any API. Fleetio's API requires proper authentication and secure communication channels.

TIPS

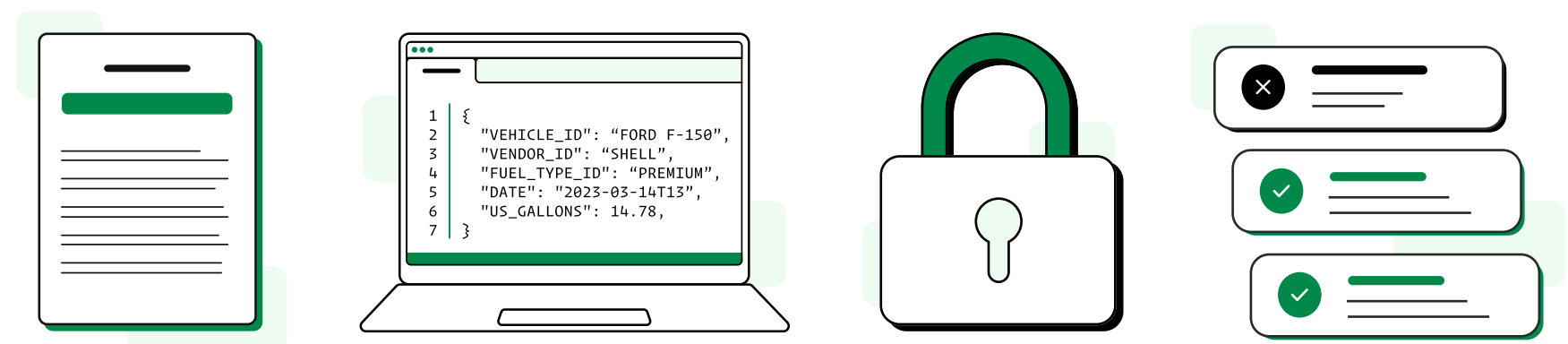
- **Use HTTPS:** Ensure all API requests are made over HTTPS to encrypt data in transit and protect it from interception.
- **Authentication:** Use API keys for secure access to the API. Store these credentials securely and never expose them in client-side code.
- **Rate Limiting:** Respect the API rate limits to avoid service disruptions.

3 Test Thoroughly

Thorough testing is important to ensure your integration works as expected and handles errors gracefully.

TIPS

- **Automated Tests:** Develop automated tests to verify API calls and responses.
- **Error Handling:** Test how your integration handles different error scenarios, such as invalid inputs or network failures.



4 Implement Error Handling

Proper error handling ensures your integration can recover from failures and provide meaningful feedback to users.

TIPS

- **Error Codes:** Familiarize yourself with common API error codes and their meanings.
- **Logging:** Log errors and API responses for debugging and monitoring purposes.
- **User Feedback:** Provide clear and actionable error messages to users.

5 Optimize Performance

Optimize your API integration for performance to ensure it runs efficiently and does not overburden your systems or Fleetio's API.

TIPS

- **Batch Requests:** Where possible, batch multiple create or update requests into a single call using our Bulk API.
- **Caching:** Implement caching strategies to minimize redundant API calls and improve response times.
- **Rate Limits:** Monitor and respect API rate limits to avoid throttling and service interruptions.



6 Monitor API Usage

Regular monitoring of your API integration helps you ensure it's performing as expected and allows you to identify and address issues proactively.

TIPS

- **API Logs:** Regularly review API logs for errors and performance metrics.
- **Usage Metrics:** Track key usage metrics such as the number of API calls, response times and error rates.
- **Alerts:** Set up alerts for unusual activity or performance issues.

7 Keep Your Integration Updated

APIs evolve over time, with new features being added and deprecated as the product changes. Keeping your integration updated ensures continued compatibility and access to our latest features.

TIPS

- **Versioning:** Pay attention to API versioning and update your integration as needed.
- **Changelog:** Regularly review Fleetio's API changelog for updates and deprecations.
- **Backward Compatibility:** Ensure your integration remains backward compatible with older API versions if necessary.

8 Collaborate with Fleetio's Support Team

Effective collaboration with Fleetio's support team can help you resolve issues quickly and optimize your integration.

TIPS

- **Documentation:** Provide clear and detailed documentation when seeking support.
- **Feedback:** Share feedback with Fleetio (help@fleetio.com) to help improve the API and its documentation. We want to know what has worked for you!

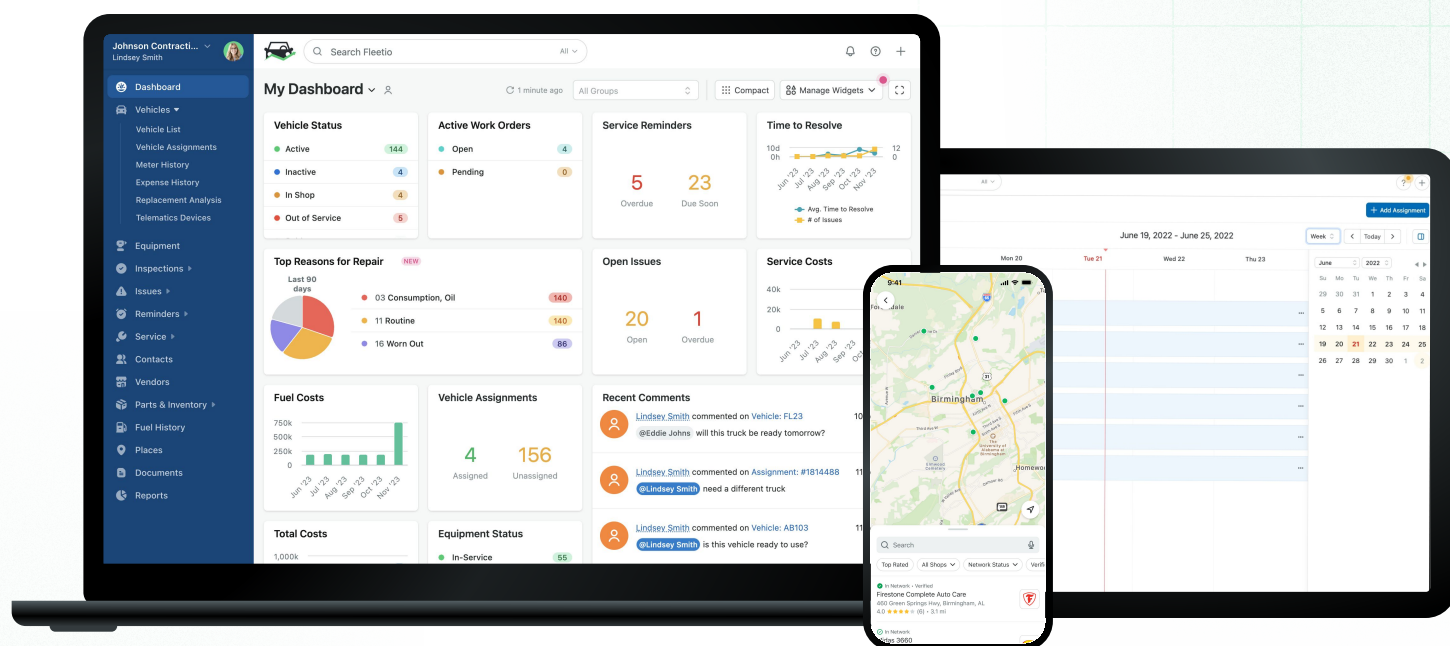
Getting Started with Fleetio's API

API Quick Start

Getting started with Fleetio's API is straightforward and only takes three simple steps:

- 1 Generate an API Key in your Fleetio account in order to securely interact with our API.
- 2 Set Account-Token HTTP Header and Authorization HTTP Header to authenticate your requests.
- 3 Put it all together and make requests to the Fleetio API using your programming tool of choice!

For more details, we offer a full guide to our API on our Developer Portal at developer.fleetio.com →



See how Fleetio can help you create an ideal data environment for your fleet.

Get a Free Demo

fleetio.com/demo